MIL-B-565A
30 SEPTEMBER 1958
SUPERSEDING
JAN–B–565
FEBRUARY 1948

MILITARY SPECIFICATION

BENZIL, TECHNICAL

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers one grade of benzil ($C_6H_5COCOC_6H_5$).

2. APPLICABLE DOCUMENTS

2.1 The following specification and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

PPP–D–723 — Drums, Fiber.

STANDARDS

MILITARY

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129 — Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply:

INTERSTATE COMMERCE COMMISSION

49 CFR 71–78 — Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and other Dangerous Articles.

(The Interstate Commerce Commission regulations are now a part of the Code of Federal Regulations (1949 Edition — Revised 1950) available from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Orders for the above publication should cite "49 CFR 71–78 (Rev. 1950).")

CONSOLIDATED CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

FSC 6810

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(Application for copies of these freight classification rules should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

3. REQUIREMENTS

3.1 **Color.** The material shall have a yellow color.

3.2 **Odor.** The material shall have an almond-like odor similar to benzaldehyde but not an acid odor similar to nitric acid.

3.3 **Chemical.** The material shall conform to the requirements of table I.

TABLE I. *Requirements*

Requirement	Minimum	Maximum	Test method
Moisture, percent	...	0.50	4.3.1
Crystallizing point, °C	93.0	95.0	4.3.2
Residue on ignition (dried material), percent	...	0.10	4.3.3
Acidity (as HCl), percent	...	0.2	4.3.4
Cyanides, percent	...	None	4.3.5
Nitrogenous impurities (as NO_3), percent	...	0.40	4.3.6
Sulfates	To pass test		4.3.7
Heavy metals	To pass test		4.3.8
Iron	To pass test		4.3.9

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 **Inspection provisions.**

4.2.1 *Lotting.* A lot shall consist of that quantity of benzil produced by one manufacturer in not more than 24 consecutive hours under essentially the same manufacturing conditions and with no change in materials, provided the operation is continuous. In the event that the process is a batch operation, each shall constitute a lot.

4.2.1.1 *Batch.* A batch is defined as that quantity of material which has been manufactured by some unit chemical process and subject to some physical mixing operation intended to make the final product substantially uniform.

4.2.2 *Sampling.*

4.2.2.1 *For examination.* Sampling shall be conducted in acordance with Standard MIL–STD–105.

4.2.2.2 *For test.* A sample shall be taken from each lot, the size to be calculated on the basis of one tenth (1/10) of the square root of the number of containers in the lot raised to the next highest whole number. If there are fewer than three containers

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in a lot, each container shal be sampled. In all other cases, no fewer than three containers shall be selected. From each container in the sample a representative one pound specimen shall be taken and placed in separate, clean, dry containers which are labeled to identify the lot and container represented.

4.2.3 *Examination.*

4.2.3.1 *Nondestructive.* Each lot shall be inspected in accordance with the classification of defects and Standard MIL–STD–105.

4.2.3.2 *Tests.* Equal portions of each specimen selected in accordance with 4.2.2.2 shall be thoroughly mixed to form a composite specimen. The composite specimen shall be subjected to the tests specified in 4.3.

4.2.3.3 *Color and odor.* The composite specimen shal be examined for conformance with 3.1 and 3.2.

4.2.3.4 *Classification of defects.*

4.2.3.4.1 *Preparation for delivery* (level A only):

Categories and defects. AQL 4-0 percent defective.

Major

1. Container damaged.
2. Container leakage.
3. Container closure incorrect.
4. Container incorrect.
5. Type of container closure incorrect.
6. Marking incorrect, missing or illegible.

4.3 Tests. Distilled water and analytical reagent grade chemicals shall be used throughout the tests. Where applicable a blank determinations shall be run and corrections applied where significant.

4.3.1 *Moisture.* Weigh rapidly to the nearest 0.1 milligram (mg.) approximately 5 grams (gm) of the specimen into a tared, covered, low-form weighing dish of such size that the specimen forms a layer approximately 0.5 centimeter deep when spread over the bottom of the dish. Remove the cover and dry the material in a vacuum desiccator over concentrated sulfuric acid or phosphorus pentoxide for 24 hours. Cover the dish, weigh to the nearest 0.1 mg., and calculate the percent moisture as follows:
where:

$$\text{Percent moisture} = \frac{100 \times (W - A)}{W}$$

W — original weight of specimen, in gm.

A — weight of specimen after drying, in gm.

Retain the dried specimen for test given in 4.3.3.

4.3.2 *Crystallizing point.* Fill a clean, dry, 1 by 6-inch test tube about three-quarters full of the specimen. Place the tube in an oil bath at a temperature approximately but no higher than 105°C. and melt the material with continuous stirring. When the temperature of the clear, molten mass reaches 99°C. (which must not require more than 15 minutes of total heating time), remove the tube from the bath at once, wipe the oil from the tube, and quickly assemble the apratus shown in figure 1. Stir continuously until crystallization starts, then discontinue stirring and watch the temperature on a thermometer which has been calibrated by the National Bureau of Standards within 2 years of the date of its use in this test. The thermometer shall be graduated at least to 0.1°C. The mass will supercool and, with crystallization, the temperature will rise to a value which will persist for at least 1 minute. Calculate the crystallizing point as follows:
where:

$$\text{Crystallizing point} = T + N \times (T - t) \times (0.000159)$$

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T = uncorrected crystallizing point.

N = number of degrees in the exposed column.

t = average temperature of the exposed column as indicated by the correction thermometer.

4.3.3 *Residue on ignition.* Weigh to nearest 0.1 mg. approximately 2 gm. of the dried specimen (see 4.3.1) into a tared crucible and add 0.5 milliliters (ml.) of concentrated sulfuric acid. Heat slowly until all the carbon is consumed. (The heating is preferably carried out in an electric muffle furnace at 600° to 700°C.). Cool in a desiccator, weigh to the nearest 0.1 mg., and calculate the percent residue on ignition as follows: where:

$$\text{Percent residue on ignition} \quad \frac{100 \times A}{W}$$

A = weight of residue, in gm.

W = weight of dried specimen, in gm.

Reserve the residue for the tests specified in 4.3.8 and 4.3.9.

4.3.4 *Acidity.* Weigh to the nearest 0.1 mg. approximately 2 gm. of the specimen and heat to boiling with 50 ml. of water under a reflux condenser. Cool, add two drops of phenolphthalein indicator, titrate with 0.1 *N* sodium hydroxide solution, and calculate the percent acidity as follows: where:

$$\text{Percent acidity (as HCl)} = \frac{3.65 \times V \times N}{W}$$

V = volume of sodium hydroxide solution used, in ml.

N = normality of sodium hydroxide solution.

W = weight of specimen, in gm.

4.3.5 *Cyanides.* Shake 1 gm. of the finely powdered specimen for 5 minutes with 10 ml. of water. Filter and add 2 ml. of yellow ammonium sulfide to the filtrate. Evaporate to 2 ml. on the steam bath, add 4 ml. of a 1 to 1 solution of hydrochloric acid and filter. To the filtrate add 1 drop of a 9-percent solution of ferric chloride solution. No orange or red color shall be formed.

4.3.6 *Nitrogenous impurities.* Place 1 gm. of specimen in a 500 ml. Kjeldahl flask and add 30 ml. of concentrated sulfuric acid containing 1 gm. of dissolved salicylic acid. Mix thoroughly and let stand for 30 minutes with frequent shakings. Add 5 gm. of powdered sodium thiosulfate, mix thoroughly, incline the flask and heat gently at first. Gradually increase the heat and boil briskly for 5 to 10 minutes. Then add two selenium granules and continue to boil until the solution has been clear for 20 to 30 minutes. Cool, carefully add 250 ml. of water, cool again and add, down the side of the flask, 100 ml. of 30-percent solution of sodium hydroxide. Add immediately a few pieces of granulated zinc and distill 150 ml. of the solution into a receiver containing 20 ml. of 0.02 *N* sulfuric acid and 30 ml. of water. The delivery tube from the condenser should dip below the surface of this solution. Titrate the excess acid in the receiver with 0.02 *N* sodium hydroxide. Calculate the percent of nitrogenous impurities as NO_3 as follows:

where:

$$\text{Percent nitrogenous impurities as } NO_3 = \frac{6.2\,(V \times N - B \times M)}{W}$$

V = volume of sulfuric acid used, in ml.

N — normality of sulfuric acid.

B = volume of sodium hydroxide solution used, in ml.

M — normality of sodium hydroxide solution.

W = weight of specimen in gm.

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4.3.7 *Sulfates.* Add 20 ml. of water to 2 gm. of the specimen and heat to boiling. Cool, filter, and wash with water to make the filtrate (conveniently collected in a graduate) measure 20 ml. To 15 ml. of the filtrate add 1 ml. of 1 *N* hydrochloric acid and 2 ml. of a 10% solution of barium chloride. Any turbidity shall be no greater than that of a control made by adding 0.3 mg. of sulfate ion (in the form of a standard potassium sulfate solution) to the remaining 5 ml. of the filtrate, diluting to 20 ml., and treating with 1 ml. of 1 *N* hydrochloric acid and 2 ml. of 10% barium chloride solution.

4.3.8 *Heavy metals.* To the residue obtained on ignition (see 4.3.3) add 3 ml. of concentrated hydrochloric acid and 1 ml. concentrated nitric acid and evaporate to dryness on the steam bath. Warm the residue with 2 ml. of 1 *N* hydrochloric acid, dilute with water to 40 ml., and divide into two 20-ml. portions. Reserve one of the portions for the iron test (4.3.9). To the other 20 ml. portion, add 10 ml. of water saturated with hydrogen sulfide. Any darkening shall be no greater than that of a control made, using the same reagents, in the same manner, and adding 0.03 mg. of lead in the form of a standard lead nitrate solution.

4.3.9 *Iron.* To the second 20 ml. portion from 4.3.8 add 1 ml. of concentrated hydrochloric acid solution and 2 ml. of 30% solution of ammonium thiocyanate. The color produced shall be not more intense than that of a color made using the same reagents, in the same manner, and adding 0.03 mg. of iron in the form of a standard ferric ammonium sulfate solution.

4.4 **Rejection.** If composite specimen selected as specified in 4.2.3.2 fails to conform to this specification, the lot shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 *Level A.* Benzil shall be packed in a 20-gage full removable head metal drum with a gasketed lever-type locking ring closure and meeting the requirements of ICC Specification 37A or in a fiber drum with a gasketed lever-type locking ring closure, meeting the requirements of type III, grade A of Specification PPP–D–723.

5.1.2 *Level B.* Not applicable.

5.1.3 *Level C.* The benzil shall be packed in accordance with the manufacturer's commercial practice to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers used shall comply with ICC Regulations, Uniform Freight Classification Regulations or other common carrier regulations applicable to the mode of transportation.

5.2 **Marking.** In addition to any special marking required by the contract or order, all shipping containers shall be marked in accordance with Standard MIL–STD–129, and ICC Regulations.

6. NOTES

6.1 **Intended use.** The benzil covered by this specification is intended for use as ingredient of protective ointment.

6.2 **Ordering data.** Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Selection of applicable level of packing.

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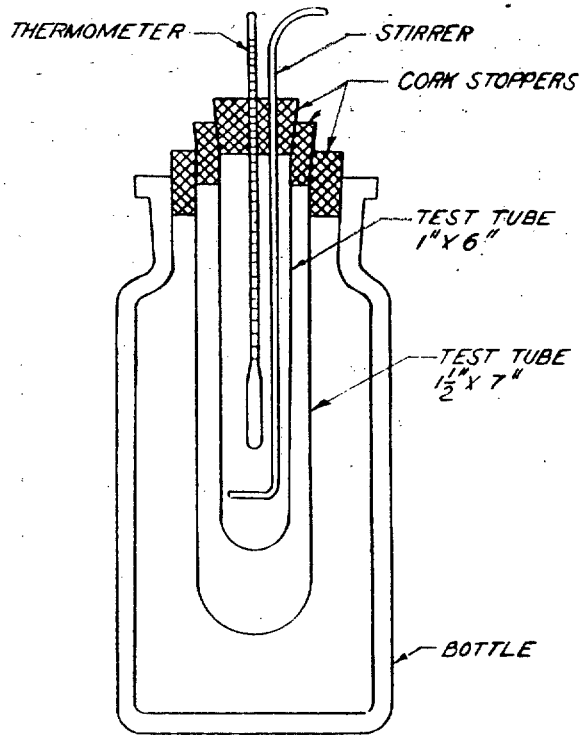


FIGURE 1. *Apparatus for determining crystallizing point.*

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